

CUSTOMER MILESTONE BRIEF

MS #38 - JCL Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.38.0
Date	2026-08-30
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #38 - JCL Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Qualifies a bounded JCL discovery and semantic subset while keeping JES, catalog, scheduler, and restart claims evidence-dependent.

Why this matters to a customer

Qualifies a bounded JCL discovery and semantic subset while keeping JES, catalog, scheduler, and restart claims evidence-dependent.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Bound JCL qualification to the canonical graph inventory of 46 jobs, two procedures, 119 steps, 451 DD allocations, and their exact graph identity.
- Added a deterministic 30-case synthetic JCL conformance corpus with 20 targeted semantic-boundary cases, 24 passing cases, six explicit fail-closed cases, and 36 observed feature categories.
- Added a 22-entry JCL compatibility ledger covering jobs, procedures, symbolics, EXEC resolution, DD allocation, DISP, GDG, DCB/SPACE, condition codes, restart, utilities, scheduler directives, JES controls, subsystem boundaries, and RACF submission identity using all five governing classes.
- Separated ten gates for estate inventory, corpus provenance, parsing, job/procedure semantics, program resolution, dataset allocation, condition/restart behavior, utilities/scheduler/security, tamper resistance, and authorized JES/catalog/scheduler

execution.

- Added frozen schemas, POSIX/PowerShell entry points, release-gate integration, deterministic receipts, exact-corpus checks, diagnostic-location checks, and rehashed-overclaim tests.
- Kept native JCL, JES, catalog/SMS, scheduler, runtime, restart, mainframe equivalence, and production readiness false; no customer or IBM source is included in the synthetic corpus.

How it differs from earlier work

MS #38 builds on MS #37 (PL/I Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds jcl qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use jcl qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept native JCL, JES, catalog/SMS, scheduler, runtime, restart, mainframe equivalence, and production readiness false; no customer or IBM source is included in the synthetic corpus.

Source of record

- CHANGELOG.md release section(s): 0.38.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.